02-24-26

CSC 659 859
Spring 2026

HW 2 –ML pipeline experiment with Random Forest ML (RF) using R or SciKit SW Toolkits

25 points
DUE: check Canvas

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HW 2 objective

Learn how to use Random Forest (RF) ML with basic SW tools (SciKit or R) and with GenAI help <u>for a full professional level RF pipeline for biomedical application</u>:

- Audit and import *Original Training Database (biomedical application)*
- Create *training and verification* DB from original DB
- Train RF and choose the optimal RF using OOB and CV methods, compute accuracy measures from each and compare
- Compute RF feature ranking and discuss explainability wrt. ground truth
- Apply run time (optimal) RF in prediction mode on verification DB
- ***Present key results in an easy to read way with full data provenance. Check, analyze and evaluate results – this is also graded and very important***

- HW 2 covers full professional level ML pipeline for classification using RF – one of the most powerful ML for tabular data
- RF is efficient so professional scale RF (as in this HW 2) can be done on a laptop (unlike deep learning)

This is great project for your CV and e-portfolio

- Data for HW 2 is be provided – it is real data from a biological study – see Canvas HW 2 info
- Helping documents for use of SciKit and RF also provided in Canvas HW 2 info
- RF tools are open and free

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Goals and Training data

Goal: classify i1 nerve cluster data from JCV (Craig Venter Inst, San Diego CA) experiment. <u>Features</u>: gene expression levels; <u>Decision classes</u>: sample is i1 cluster or not

- Data is described in paper covering biology background of **i1** data and its ground truth (<u>how i1 is defined</u>) – posted on Canvas
  - Aevermann B., Novotny M., Bakken T., Miller J., Diehl A., Osumi-Sutherland D., Lasken R., Lein E., Scheuermann R.: "Cell type discovery using single cell transcriptomics: implications for ontological representation", Human Molecular Genetics 27(R1): R40-R47 · March 2018 (provided in Canvas)
- Data for HW 2 is posted on Canvas: "***Original training DB i1 cluster.csv***"
  - *Above* XLS file has ***609 columns with 608 features*** e.g. values of gene expressions, and **last column called *"label"*** as class label, with ***value of 1 indicating + class sample (i1 cluster),*** and value of ***0 indicating – (non i1 class) class sample***
  - **Training data have only numerical features (columns), no missing data (0 for feature value mean's value of 0)**

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i1 cluster ground truth

- i1 cluster data from JCV Paper describing biology background of i1 data and its ground truth (how cluster i1 is defined) in paper below - no need to read the paper, it is for reference
  - Aevermann B., Novotny M., Bakken T., Miller J., Diehl A., Osumi-Sutherland D., Lasken R., Lein E., Scheuermann R.: "Cell type discovery using single cell transcriptomics: implications for ontological representation", Human Molecular Genetics 27(R1): R40-R47 · March 2018 (provided in Canvas)
- Ground truth for i1 cluster is established by independent biological research – the paper above established what is i1 cluster AND what genes indicate and activate for creation of i1 ➔ important to compare with your feature ranking later:
- ***<u>i1 cluster</u>: A human MTG cortical layer 1 GABAergic interneuron that selectively expresses COL5A2 and NDNF and FAT1 mRNAs***

Note: in the feature list there are families of COL5A2 genes - they use same name prefix – consider them the same genes for feature ranking analysis purposes

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HW 2 report outline — MUST follow the outline and section titles (1) (details in subsequent slides)

1. **Title page** (HW 2 title, class, semester, date, your name, your email)
2. **Audit of Original database**: describe source of data, and audit the DB ***following guidance in class slides***. Provide all data stats
3. **Creation of *training DB* and *verification DB*:** Explain how you removed two random samples (one class 1 one class 0) and then summarize the stats of the remaining Training DB (original DB minus two random samples) and Verification DB (2 random samples)
4. **SW tools**: explain what tools you used, for the HW
5. **Experimental Methods and Setup**
   1. Describe Method 1: basic training with OOB accuracy estimate. What ntree, mtry, cutoff ranges you used for RF training (e.g. for grid search). Show full set of accuracy measures you will extract. Describe SW methods/calls/classes
   2. Describe Method 2: Using 3 fold CV describe steps for each fold. ***Can not use ready made CV methods, must do each fold manually.*** Describe SW methods/calls/classes
   3. Describe Verification and comparison of results from Methods 1 and 2

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Use page size as needed

HW 2 report sections (2)

6. **Actual Results of RF Training and Accuracy Estimates**: Show Method 1 results then Method 2 results. Then compare results. Show key code snippets and describe ALL relevant data in easy to read way (tabular preferred) with full figure/table captions
7. **Feature Ranking**: Explain how you obtained best trained RF. Show feature ranking for top 10 ranged features (use MDA for R, GINI method for SciKit). Show code snippets. **Describe**: How does it help you in explaining how RF worked? What did you learn doing this step? How it compares with ground truth established in biology paper for i1 cluster?
8. **RF Run Time test**: Create RF run time engine (best trained model). Take 1 positive and 1 negative sample from verification database and run them through Run Time RF. to predict their class and probability of predicted lasses. Show code snippets and tool output and classification results and compare prediction with ground truth. **Describe**: What is your confidence in RF predictions and based on what you determine it? Show RF output as class decision and class probability
9. **Resources:** List resources/references, what GenAI you used, and names of students who helped you (if any)

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HW 2 report sections (3)

- **Appendix I:** if you used GenAI explain how you used it with examples of your interaction and its output 1-2 pages. For each task you used GenAI:
  - Describe task for which you used GenAI
  - Show example prompts
  - Show key GenAI output
  - Rank usefulness from 1…5, 5 being very useful
- **Appendix II**: Show full code (copy the code into PDF format – do not send executable) . Document the code please (this will be your portfolio). (In main body of the report show only key code snippets).

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HW 2 Report – <u>one</u> PDF

HW 2 PDF file consists of:

- **Title page**
- **Main body (sections 2-8, as explained before):**
  - *Report writing guidance (important, will be graded)*
    - Have <u>separate</u> sections titled and ordered exactly as provided in previous slides **More details later**
    - Present key/summary data and results in tables or graphs. Graphs and table can be cut and pasted from tools or can use R Markdown or Jupyter Notebook. Each table/chart has to have caption and full provenance: description of <u>all parameters and settings and data used</u>
    - Show <u>main key snippets of c</u>ode in main body where necessary for easy reading
- **Resources (sec. 9):** list references used, and list who helped you
- **Appendixes I and II** (code in PDF please for security)

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More guidance on HW 2 sections

- **MUST follow**

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<u>Section 2</u> – Original Database audit

- ***Summarize*** all important data statistics (number of samples, classes, features….) for your training database (original data minus 2 verification samples) and *verification database* in an easy to read way (<u>table format preferred</u>)
- ***Audit the training database***: *Use class slides on the subject* as a checklist and use info in HW 2 slides (see next slide – use it as a template for audit)
  - Provide quoted biology paper as a reference to data source and use info provided in preceding slides

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Important: *Training DB audit Checklist*– **MUST do in every application (also in team project)**

1. Description of features and data formats are well documented
2. How are class labels obtained/verified wrt. ground truth

   (For 1 and 2 say it briefly and then refer to research paper)
3. Is demography well covered in adequate and fair way
4. Check number of samples in each decision class: is some class **unbalanced** (unbalanced class is one having less than 10% of all class samples)
5. Type of each feature (numerical, categorical nominal or ordinal) – needs to be declared in SW tools
6. Are there missing values and do they need to be imputed and how
7. Are there enough samples compared to number of features used (must be at least 10 X more samples than features)
8. Check privacy issues (no personally identifiable features)

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<u>Section 3</u> – Split Original Database (i1 spreadsheet in Canvas) into *Training DB* and *Verification DB*

- From original DB pick one random sample of class 1 and one random sample of class 0 and place them in <u>*Verification DB*</u> (make it in a separate file)
- Original DB minus 2 random samples above = <u>*Training DB*</u>
- <u>Summarize</u> the data statistics for Training and Verification DB (number of samples, classes, features….) in separate tables

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<u>Section 4</u> - SW tools used

- Just list main ML tools you used

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<u>Section 5</u> - Experimental Methods and Setup

Explain only what <u>you will do and how (no results here)</u>

- ***Method 1: OOB accuracy estimation***
  - Explain the method briefly
  - What specific tool and SW classes/methods you will use
  - List <u>all</u> the parameters and ranges of hyper-parameters you will use
  - How will you estimate OOB and other accuracy measures – show formulae (MUST do all we covered in class)
- ***Method 2: CV error estimation***
  - How will you do 3 fold CV - explain in max one page – must do it manually in 3 steps. ***Can not use one-liner methods for CV provided by toolkits.*** What accuracy measures will you compute in each CV fold and for final CV accuracy? (basic accuracy is OK, better yet to also do F1). What specific tool and SW classes/methods you will use
- ***How will you compare Method 1 and 2 results***

NOTE: In real life this step would be verified and approved by experts before you embark on the experiment

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<u>Section 6</u> - Actual Results of RF Training and Accuracy Estimates

Have <u>three subsections</u> devoted separately to: Method 1; Method 2; and comparison, as described in section 5. <u>Show actual results you got</u>. Follow what you proposed in section 5

- ***For Method 1 show in easy to read way:***
  - *All hyper-parameter ranges*
  - *Key code snippets focusing on ML part*
  - *Confusion matrix*
  - *All accuracy measures*

Make it all easy to read: Use tables and charts with full enumerated captions

- ***For Method 2 show un easy to read way:***
  - *For each of 3 CV folds show how many samples of class 1 and class 0 in train and test partition*
  - *Show accuracy and preferably F1 for each of 3 folds*
  - *Optional but highly recommended: show confusion matrix from test data in each CV fold*
  - *Show how you computed final accuracy and F1 and provide results. Optional but highly recommended: show final confusion matrix*
  - *Show key code snippets*

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<u>Section 6</u> – some writing guidance (2)

- From your tool output use the data to show all accuracy results (<u>confusion matrix, accuracy, recall, precision, F1 score</u>) and embed in the report – cut paste or use R Markdown or Jupyter (see later). Summarize in table
- All accuracy measures to be in 5 decimal accuracy
- MUST add text describing what you have done and how you computed it– describe all relevant parameters
- Format it nicely to make it easy to read. Use tables and charts for all key info, <u>with captions providing all the data used for the table/chart</u>
- Confusion matrix shall show full sample counts and <u>not normalized</u> ones. and axes shall be defined (*true vs. classified*)

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<u>Section 7</u> - Feature Ranking

Important: need to do feature ranking <u>from best trained model only</u>

- For R users use MDA, for SciKit users use Gini (see "hints for users" memos on Canvas for R and SciKit)
- Show key code snippet

- **Print top 10 ranked features** (for best trained RF) (cut and paste from tools or use R Markdown or Jupyter) – plot feature names and rank values and graph - Show in easy to read table
- **Discuss and comment:**
  - Do highly ranked features correspond to biological knowledge as in source paper
    - ***<u>i1 cluster</u>: A human MTG cortical layer 1 GABAergic interneuron that selectively expresses COL5A2 and NDNF and FAT1 mRNAs***
  - Do ranking values show there is obvious cluster of high raked features that dominates ranking?
  - How many top ranked feature would you suggest be used in production (tradeoff between smaller number of features used vs. accuracy reduction)?

- **Note**: in your feature rank list there may be families of COL5A2 genes (they use same prefix in the name – consider them the same genes for feature ranking analysis purposes)

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<u>Section 8</u> - RF Run Time test

- ***Create RF run time engine***
  - Describe how you did it,
  - Show key hyper-parameters and code snippet
- Run it in RF ***predict*** mode using verification database (2 samples)
  - Show classification results as class decision and probability for each class – use table format
  - Show key code snippet
- **Discuss (**in text)
  - Are verification samples correctly classified
  - Do you have confidence in given classification, what is it for each sample and how did you establish it

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<u>Section 9</u> – References and resources

- List all papers and tools used
- List names of people who helped you

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Report writing (again) – this is graded too

- Your report has to be self contained and easy to read for average ML professional

Remember data provenance

- **Follow prescribed HW 2 document structure precisely**
- Each chart/result must be fully explained (in text and in captions) by describing what it is about, listing all data and parameters used, explaining the meaning of X and Y axes etc.
- Show ALL outputs and ALL parameters e.g.: ntree, mtry, cutoff (for scikit it is 0.5 fixed), misclassification matrix (label matrix fields); OOB, accuracy; F1 score. Clearly mark and present these parameters. Show all accuracy measures with minimum 5 decimal places, do not round off
- For feature ranking say what measure you used and provide list of top 10 ranked features and their rank values in the table or plot it for better visualization (optional)
- Must be formatted and presented well for easy reading. Make sure report contains enough textual explanations for each step. **Use tabular format or charts for summarizing all key points and data with full caption info**
- Final report must be one PDF (can also export Jupyter or R Markdown into PDF)
- In Appendix II have copy of the code in PDF format (so it does not cause security issues if in executable format). Document the code with header (purpose, HW 2, your name) and in-line for major sections of code

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More on HW 2 report writing and format

- ***Writing professional reports on ML work is one objective of the class and HW 2 and is important for your CV and portfolio***
- ***Proper report writing an formatting is graded (see grading)***

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Can also present experimental results using Jupyter or R Markdown

- Use of these tools is good (**not mandatory**)
  - https://rmarkdown.rstudio.com/
  - https://realpython.com/jupyter-notebook-introduction/

BUT….

- Make sure there is enough text as required by HW 2 for each step
- Cleary mark and annotate all charts and captions
- Show key code snippets
- Must export tool outpouts into PDF – **HW 2 is in one PDF**

- (If you do not have time to learn how to use these tools then simply cut and paste code snippets and results charts into PDF document but later take time to learn them, good thing to know these)

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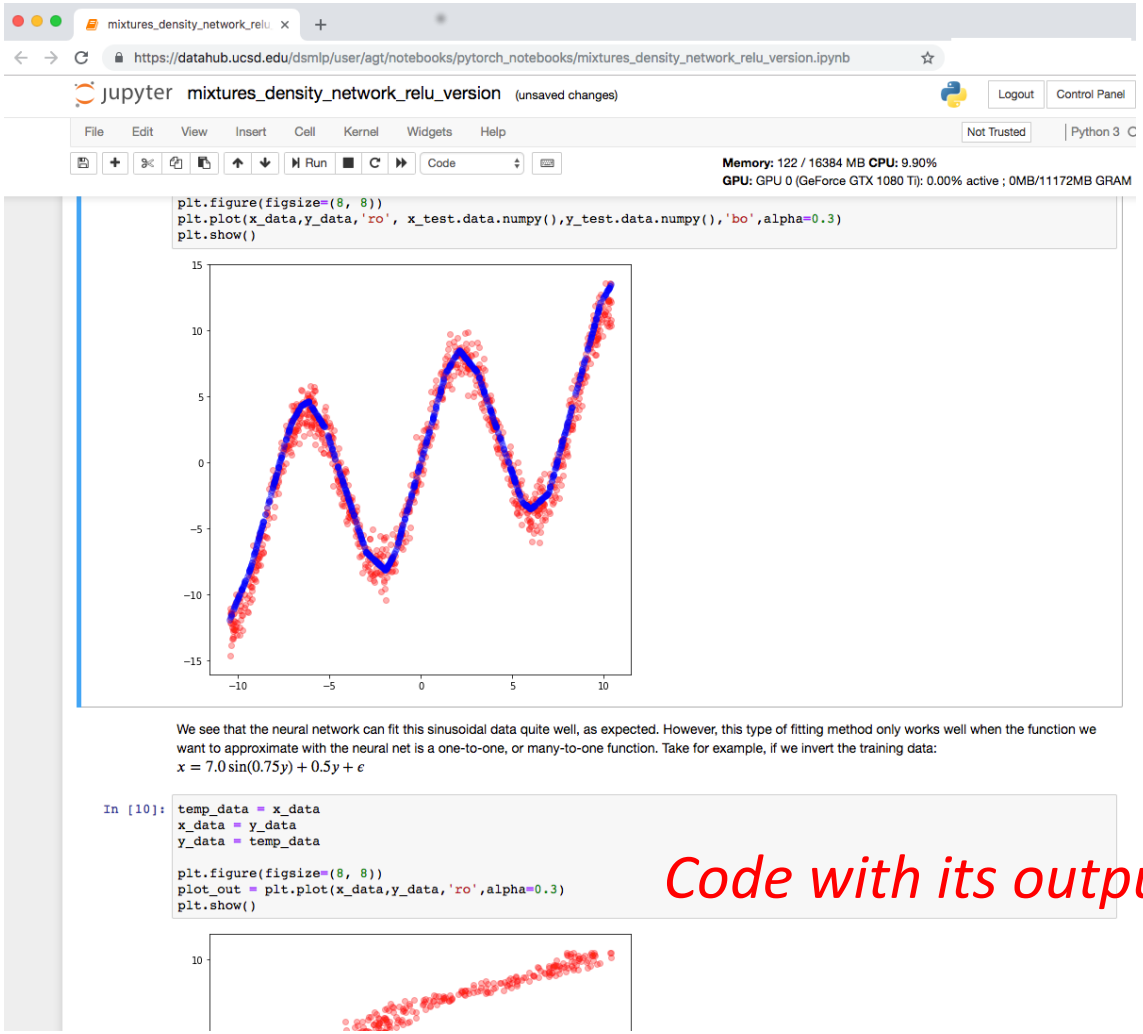
Example of Jupyter notebook

https://blink.ucsd.edu/faculty/instruction/tech-guide/dsmlp/index.html



Plain text

Code with its output

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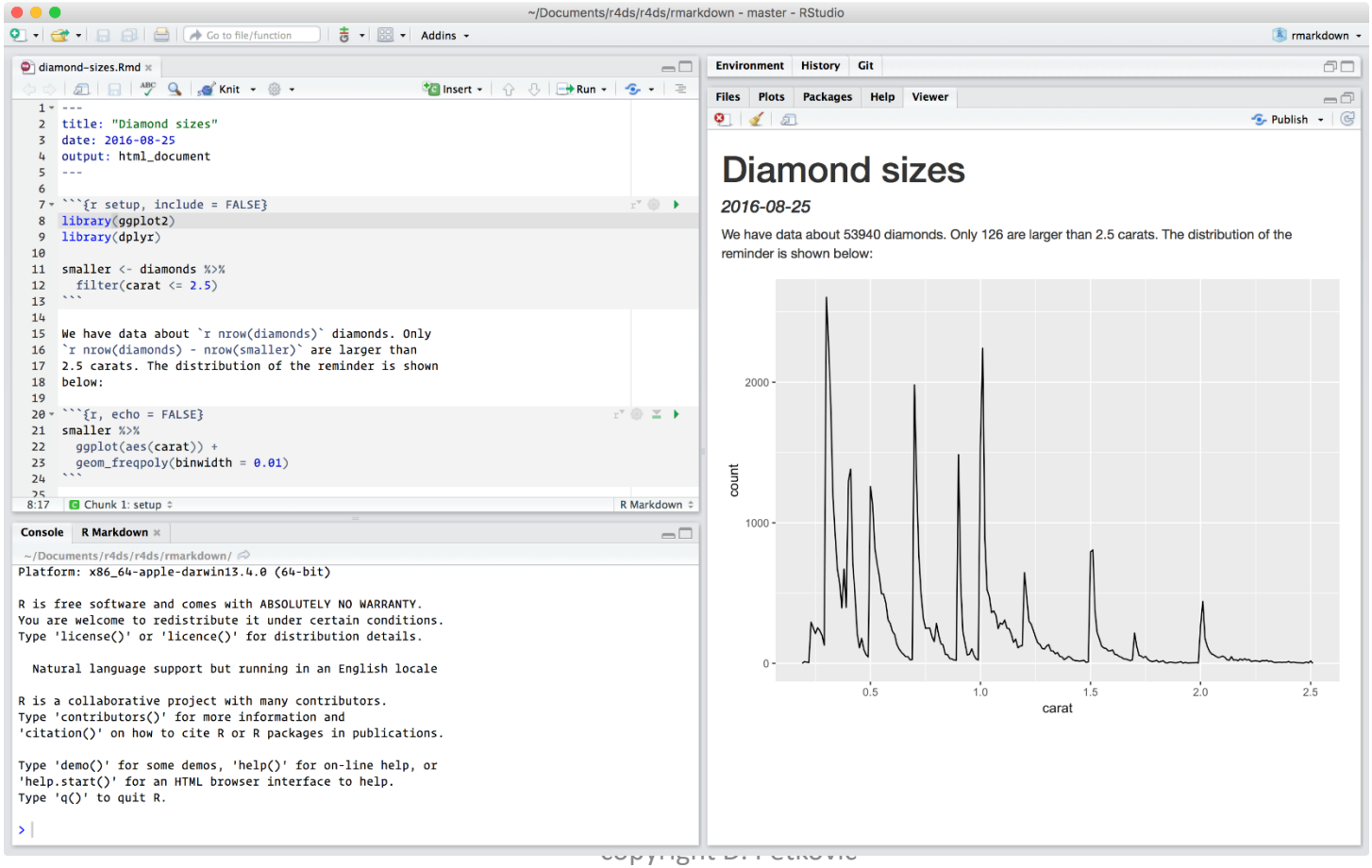
Same with R Markdown

- https://r4ds.had.co.nz/r-markdown.html



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Appendix I – GenAI use (1)

Recommendation for GenAI usage (strongly encouraged to use):

- Use it as adviser and tutor but MUST CHECK its accuracy, YOU are responsible
- One approach: Try to develop code yourself then let GenAI comment and check/correct (this way you keep your own structure and variable names)
- OK to use GenAI tool code snippets
- Make sure variable names correspond to common knowledge like MTRY, NTRYY, CUTOOFF, F1 , OOB etc. (say it in your prompts)
- **(BUT also use documents provided from instructor with code examples ("hints…" – on Canvas HW 2 info))**

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<u>Appendix I</u> – for those who used GenAI (2)

What to explain in Appendix I

- What version of GenAI tool you used
- List for all specific tasks you used it
- For each listed task say:
  - How the tool helped and how did you use it (max ½ page text but also desirable to use examples like ChatGPT output – text, code)
  - Rank how much the tool helped (1…5, 5 is best)
  - List major prompts you used

This will also be helpful for all of us to learn and we will ask volunteers to share their experience

Grading of this item is more based on how well you explained and discussed it then on how you used it

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Appendix II - code

- Code submission in PDF (cut and paste the code) – does not violate security checks
- Important to show key short code snippets in report main body, in Appendix must show full code
- Must use some text to document the code specifically related to ML issues. Don't forget header comments: purpose of the code, what it does, who coded it, date
- **Note**: this will become part of your e-portfolio for job search so this is important

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HW 2 rules…

- HW 2 is open book, open internet, ***individual work NOT team work***
  - **Can ask help from others, they can also sit next to you while you are coding (e.g. pair programming) BUT HW 2 is to be your own work – no code or text copy from other students "as is" is allowed and will be checked.**
- OK to use GenAI but YOU must check for accuracy and describe how you used it in Appendix I
- OK to use code snipes from GenAI or documents provided by instructor
- GenAI usage NOT graded as long as it is explained reasonably well
- Reference all resources you used and list names of those who helped you
- English, grammar, spelling NOT graded (but make it reasonable)
- *You might be asked to explain your code to instructor in person*

HW 2 submission

- **DUE as announced on Canvas**
- **If extension needed must ask via e-mail before the deadline**
- **Late HW submission with no approved delay -4 points**

Submsson process is also graded

- **Submission:** One PDF file with name "CSC 659-859 Spring 2026 HW 2 <your last name>.PDF" send as attachment to email to Petkovic@sfsu.edu with e-mail subject line :
  - "CS 659 859 Spring 2026 HW 2 <your last name>"
  - E-mail body has to have some short courtesy text

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Resources for HW 2, all on Canvas HW 2 section

- HW 2 Instruction slides with some useful links
- i1 data as XLS (structure explained in above slides)
- “Useful hints…” for R ad SciKit users – two memos outlying important advice, code snippets and common mistake/issues to avoid in using these tools (one for SciKit one for R users)
- Some tutorial slides from previous students of D.P.
- Also consult class slides on RF, e.g. for DB audit steps
- OK to ask GenAI (but describe it and analyze its usefulness)
- ***Ok to ask your team mates BUT YOU are to do HW 2 on your own***

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HW 2 grading – total 25 points (1)

1. **Audit of Original database: 2 points**
2. **Creation of *training DB* and *verification DB*: 1 point**
3. **SW tools**: not graded
4. **Experimental Methods and Setup: 5 points**
6. **Actual Results of RF Training and Accuracy Estimates: 8 points**
7. **Feature Ranking: 3 points**
8. **RF Run Time test: 2 points**
9. **HW 2 presentation, organization, following the required outline, section titles and organization, proper submission as per guidance provided: 4 points *(English and spelling NOT graded)***

Appendices and GenAI use summary (if used) must be provided but are not graded

TOTAL 25 points

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<u>Reminder</u> (also due to poor template and report organization in HW 1

- Follow up specified formats and section titles
- Follow structure and information organization for each section as specified
- Have all major results in a table or chart, with enumerated captions showing ALL relevant information
- **Failure to abide by these requirements will result in significant negative points**

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This HW might be challenging BUT….

- It will improve your understanding of ML and build confidence
- Will add valuable skills
- You will also practice writing of professional and scientific ML reports
- You will learn how to leverage GenAI

- ***HW 2 will help update your CV with marketable skills***

- ***Make HW 2 part of your portfolio for job search***

- ***Please do not distribute or post HW 2 anywhere except in your own github***

Have fun!!!!!!!!!!!!

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